

Aman Pandey Sanjeev Kumar

Full-time Availability: Starting May 2026

+1 (765)-543-8366 sanjeevk@purdue.edu [linkedin.com/aman-pandey-s](https://www.linkedin.com/aman-pandey-s) github.com/AmanPandey28

Education

PURDUE UNIVERSITY

Master of Science in Electrical and Computer Engineering

Specialization: Artificial Intelligence & Machine Learning

West Lafayette, Indiana

May 2026

Master of Science in Industrial Engineering

Specialization: Operations Research

- ★ **Relevant Coursework:** Machine Learning Systems, Deep Learning, Reinforcement Learning, AI Hardware, Linear Programming, Linear Algebra, Advanced Probability

Technical Skills

AI & ML Systems: Inference Optimization, Quantization, Pruning, Knowledge Distillation, Mixture of Experts, Sparsity, Latency / Throughput / P99 Analysis.

Programming: Python, C, C++, Triton, CUDA, CuTe, CUTLASS

Frameworks & Tooling: PyTorch, ONNX, OpenCV, Hugging Face, scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Nsight Compute, Nsight Systems, Docker, Kubernetes, Slurm, Git, Linux.

Systems & Compilers: MLIR, TVM, IREE, TensorRT-LLM, vLLM, SGLang, FlashInfer, KV-Cache Optimization, Distributed KV Routing, Disaggregated Inference, Request Scheduling / Batching,

Domains: Machine Learning Systems, Production AI, Inference Optimization, AI Hardware, High-Performance Computing

Projects

Latency-Constrained Image to Text Retrieval on Qualcomm XR2 Gen 2 (LPCVC 2026) | *Competition*

- ★ Built an **INT8 dual-encoder retrieval system** for edge inference under strict latency constraints on **Qualcomm XR2 Gen 2**.
- ★ Implemented a production deployment pipeline: **PyTorch** → **ONNX** → **QNN** using Qualcomm AI Hub.
- ★ Compressed the text encoder using **CLIP distillation**, preserving **Recall@10** while beating the OpenCLIP latency baseline.
- ★ Applied **graph-level optimizations** (operator fusion, norm folding, embedding reduction) to avoid CPU fallbacks.
- ★ Profiled on-device inference to iteratively refine model structure and **quantization strategy**.

Cache-Aware Routing for Distributed LLM Inference | *GitHub*

- ★ Designed a **stateful KV-cache-aware router** for multi-node LLM serving to reduce **Time-To-First-Token (TTFT)**.
- ★ Proposed a **queue-vs-prefill cost model** balancing cache locality against queueing delay under bursty traffic.
- ★ Built a **vLLM-faithful simulator** modeling paged KV caches, radix-tree reuse, eviction, and continuous batching.
- ★ Showed that cache-maximizing routing causes **request herding and P99 blowups**, while cost-aware routing stabilizes latency.
- ★ Evaluated on **multi-turn and RAG workloads (SQuAD)**, achieving stable TTFT and throughput without KV transfer.

Optimizing GPT-2 with DoRA, GQA, and Metal Backend | *GitHub*

- ★ Co-developed **Terminator**, a pure C implementation of GPT-2 integrating **DoRA** and **GQA** for efficient training and inference.
- ★ Implemented magnitude-direction decomposition, low-rank updates, and **KV head sharing** to reduce memory footprint.
- ★ Demonstrated GPT-2 on QEMU-emulated Raspberry Pi with **60–70% memory savings**.

Sparcellerate: Compiler Support for Arbitrary Sparse Tensor Formats | *GitHub*

- ★ Extended the **MLIR SparseTensor dialect** to support block-sparse, N:M, ELLPACK, and custom encodings.
- ★ Implemented a **pattern-matching pipeline** to route ops to optimized GPU kernels or a custom backend.
- ★ Developed a **Triton-based fallback codegen path** for tiled, coalesced execution under irregular sparsity.

Experience

Teaching Assistant, IE 332: Computing in Industrial Engineering

Purdue University

Aug 2023 – Present

West Lafayette, Indiana

- ★ Led lab sessions for 75 students on Machine Learning algorithms, data pipelines, and supervised/unsupervised learning.
- ★ Taught relational database design, ERDs, SQL, and backend web development (HTML/CSS/JS/PHP/MySQL).
- ★ Guided student projects integrating ML with ERP, CRM, and SCM systems.

Research Assistant

Purdue University | *GitHub*

Oct 2024 – Jan 2025

West Lafayette, Indiana

- ★ Validated clean-priority learning theory in neural networks by showing early preference for clean over noisy labels.
- ★ Analyzed test error curves and sample-wise gradient trends under varying label noise.
- ★ Ran experiments on MNIST and CIFAR-10 with CNNs, FCNs, and ResNets to study learning behavior.
- ★ Empirically supported clean-priority learning theory by demonstrating early preference for clean over noisy labels.

Data Science Researcher

Cisco X Purdue Data Mine

Jan 2024 – May 2024

West Lafayette, Indiana

- ★ Built forecast selection & ensembling algorithms for 10k+ Cisco products, improving cost-weighted prediction accuracy by **3%**.
- ★ Clustered PLIDs using **Gaussian Mixture Models** to refine forecast segmentation.
- ★ Applied Bayesian MCMC, MLE, and advanced calibration methods to optimize ensemble weighting.